

How does darkness affect pedestrian and cyclist counts in Bristol?

Assessing whether darkness discourages nighttime activity

Group members: Sharvash, Karan, Will, Tom, Jayden



Why study darkness in Bristol?



Carbon neutral by 2030 + 55% active travel ^[3]

£12m

LED upgrade

2023 - 2025 ^[1]

7% walking increase
22% cycling increase

Across East Bristol (2025) ^[2]

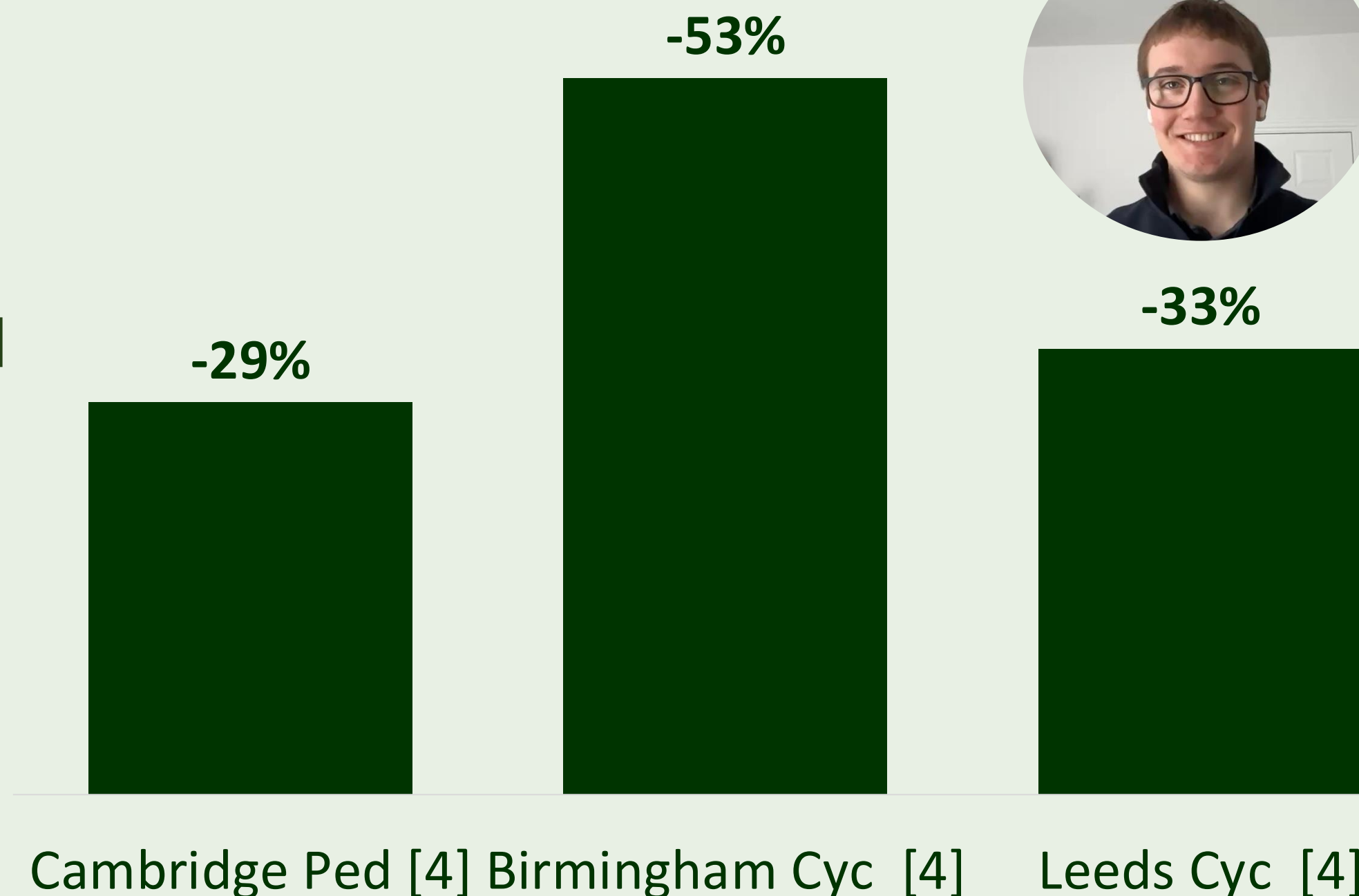
[1] Bristol LED street lighting: bristol.gov.uk
[2] East Bristol Liveable Neighbourhood monitoring report
[3] Bristol climate targets: bristol.gov.uk

What are our objectives?

1. Gather local data

2. Compare case-control OR and time-series GLM

3. Inform lighting & active-travel decisions



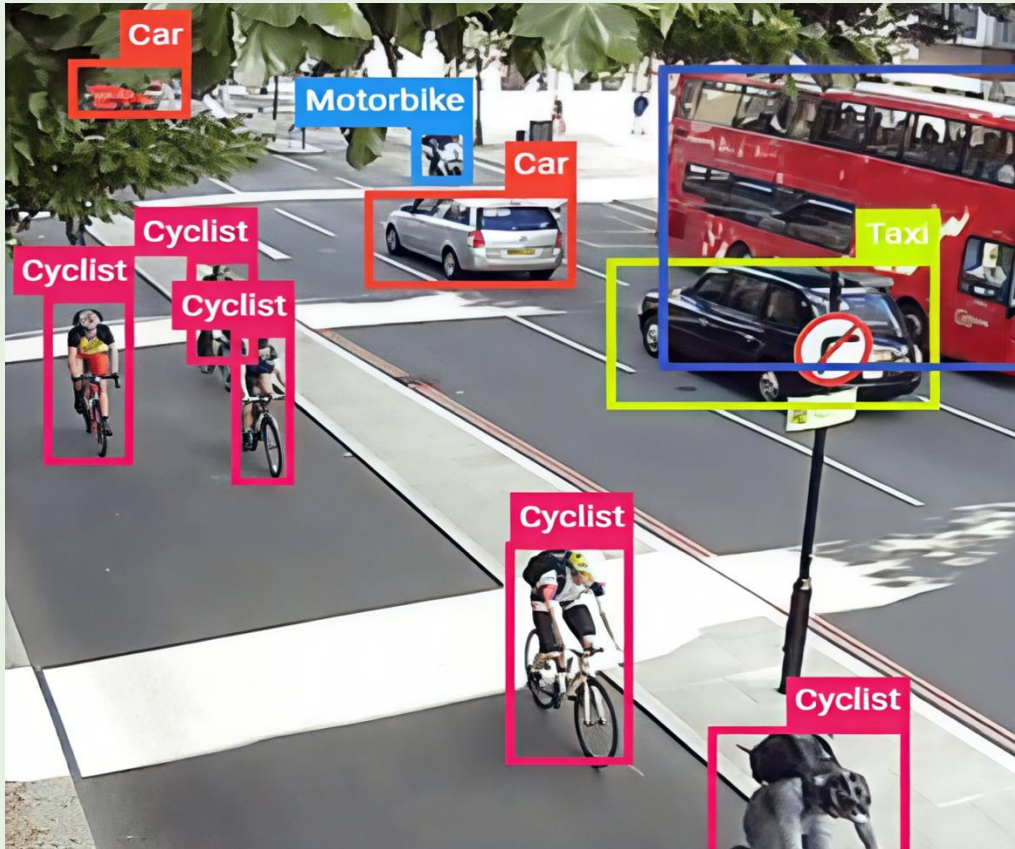
Our Data Source

1 What?



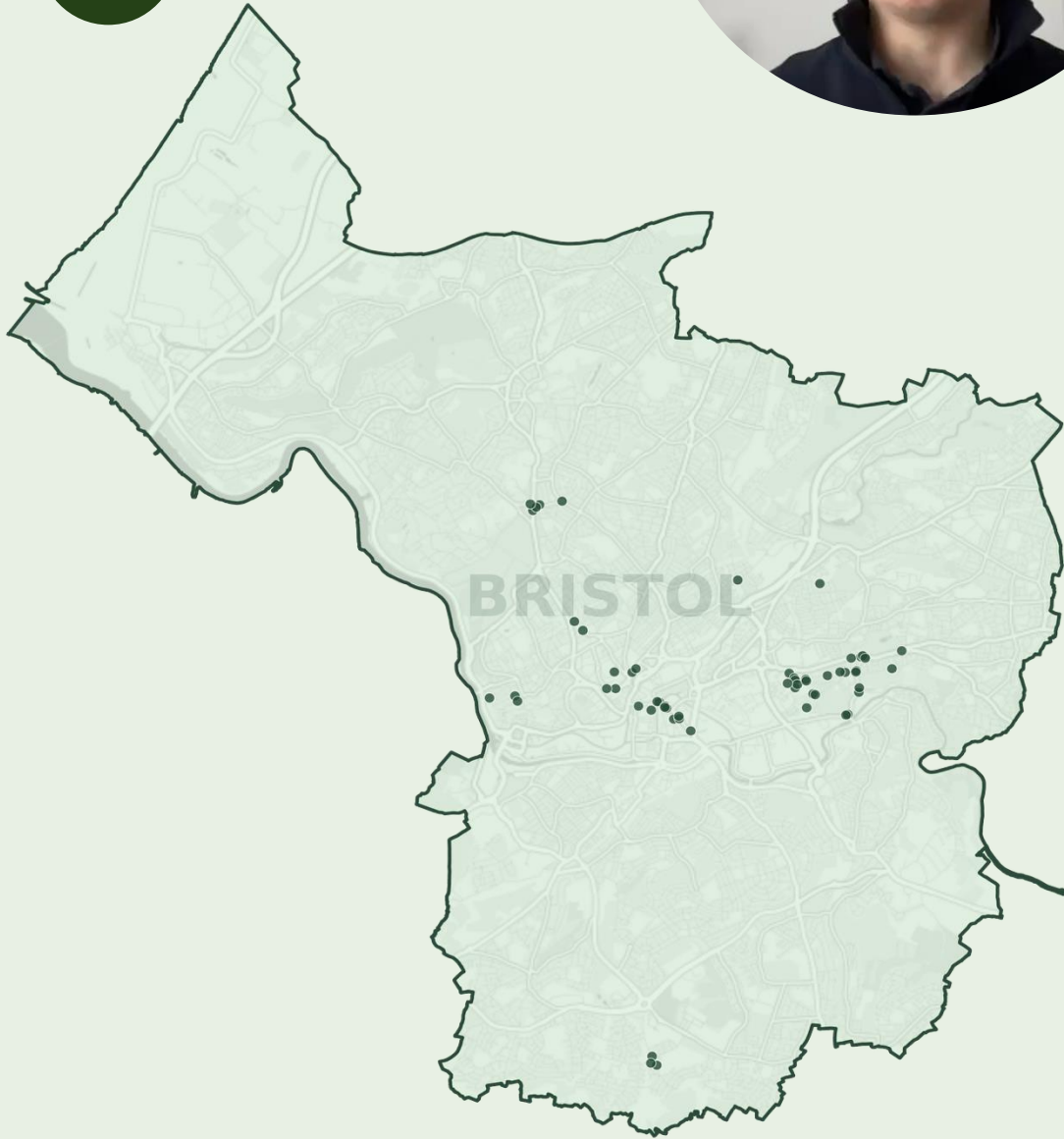
Vivacity's camera sensors [5]

2 How?



AI classification

3 Where?



58 deployed across Bristol [6]



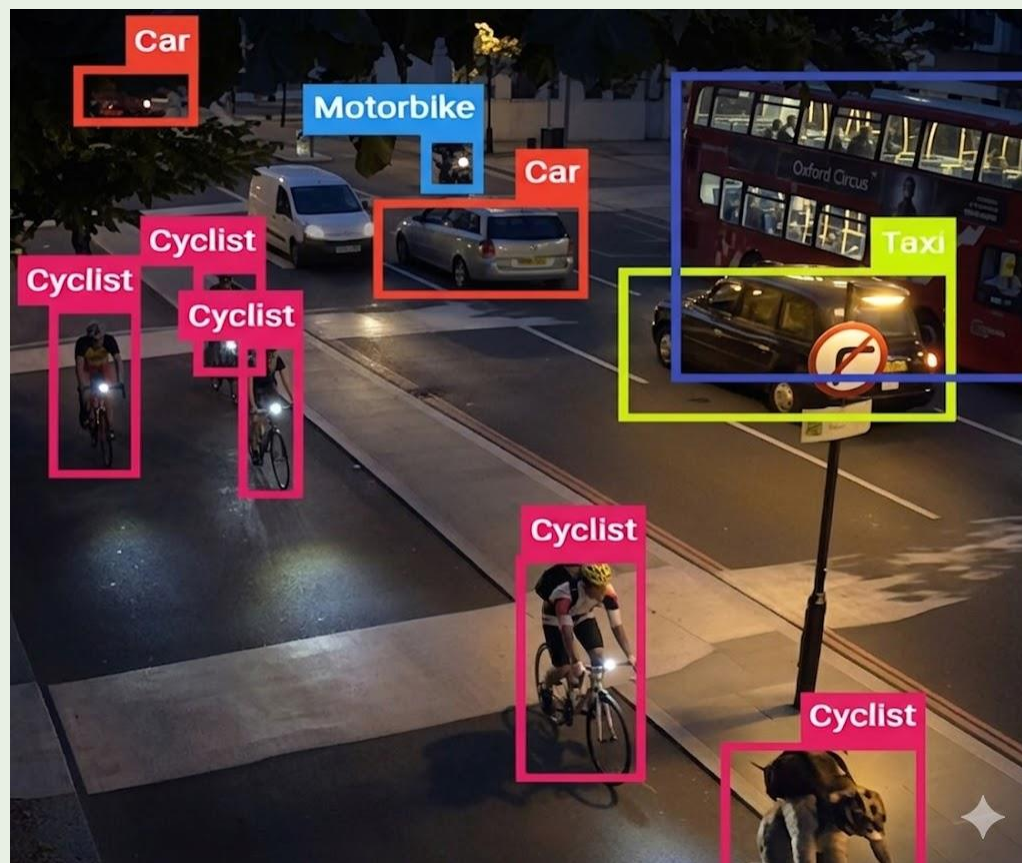
[5] Vivacity Labs, *Sensor Platform Technical Brochure*
[6] Tong & Bode, *Hourly Traffic Counts...* Bristol, UK (Zenodo).

Limitations of the Data

Accuracy of the sensor counts are not 100% accurate^[7]

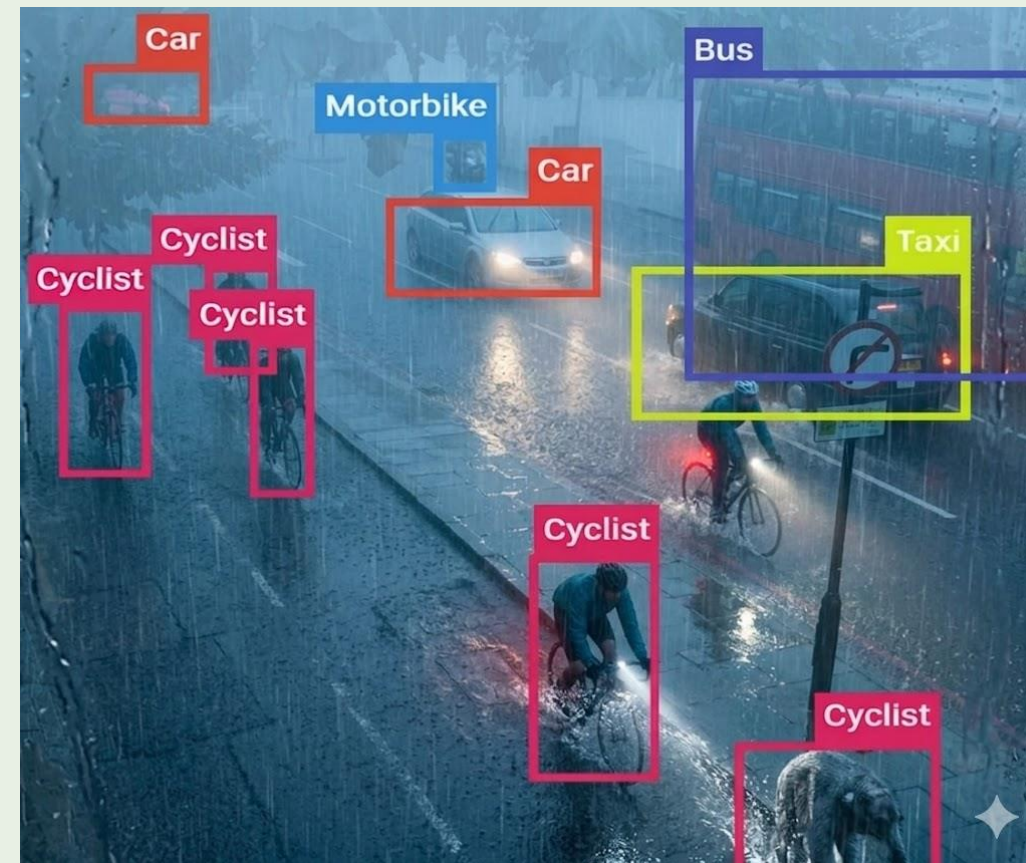
1

Darkness



2

Rain/Fog



3

Crowd Density



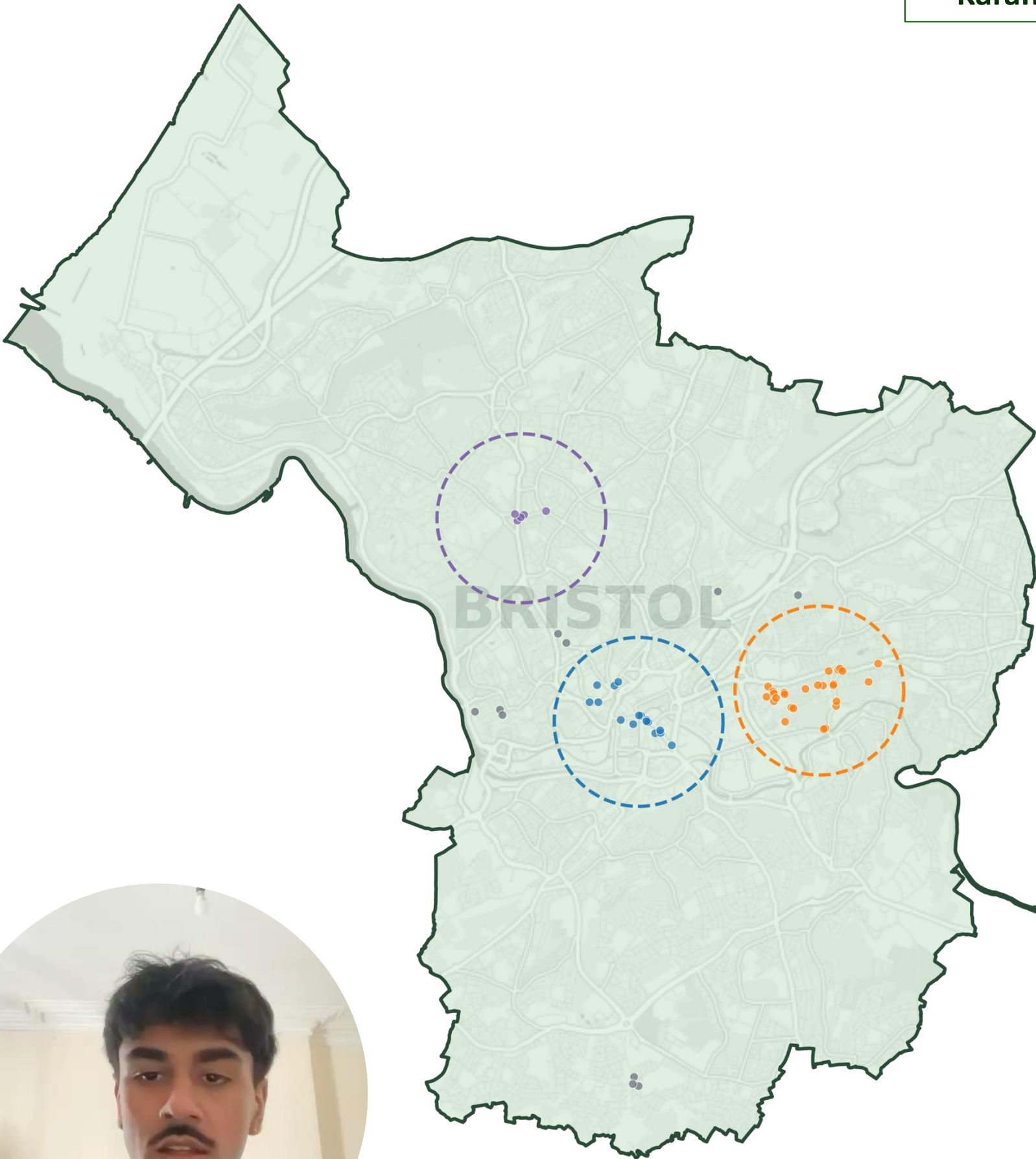
[7] Cambridgeshire County Council Research Group. (2022). Vivacity Sensor Performance Monitoring: June 2022 data.

Data Processing

Grouping Sensors by Location

Cluster	Sensors
Central	16
East	27
North	5
Outlier	10

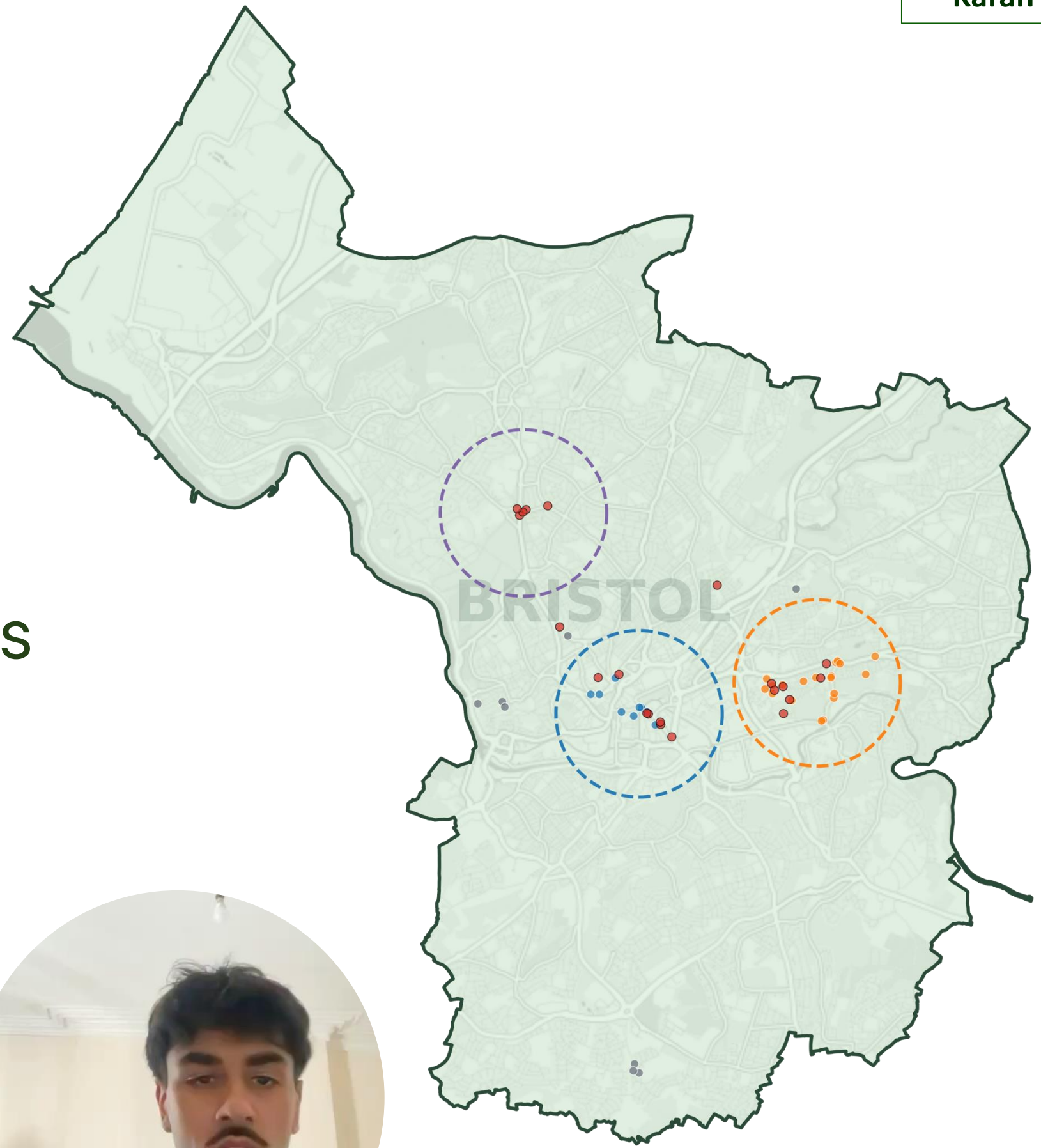
See Technical note PDF for detail on how we did this.



Data Processing

Addressing Missing Values

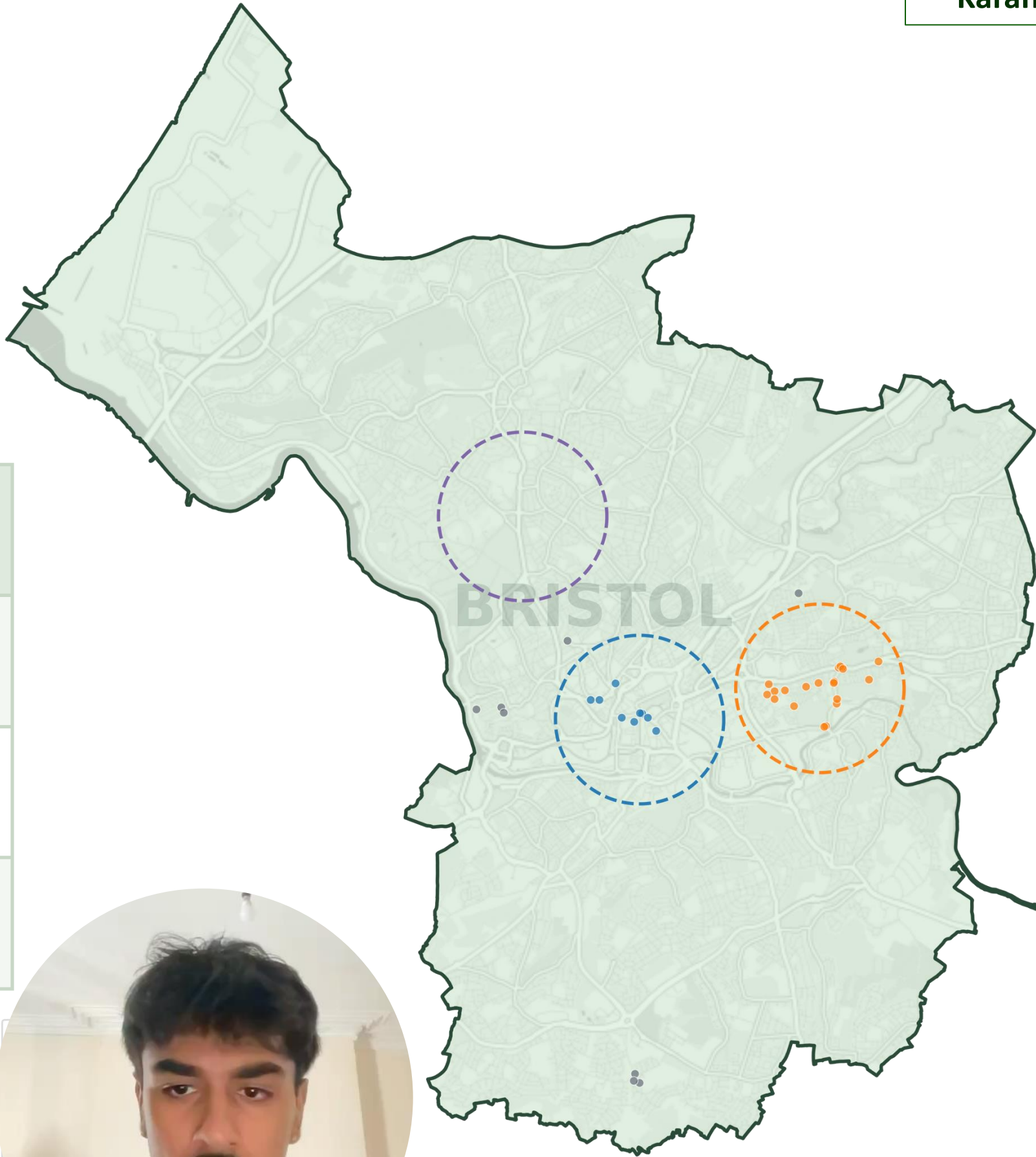
- Removed sensors with 7+ consecutive zero-count days across pedestrians, and cyclists.
- Reduced bias
- Increased accuracy



Data Processing

Addressing Missing Values

Metric	Before	After
Sensors	58	37
Hourly rows	509,472	325,008
North cluster sensors	5	0



Case-Control Method

Case-control design and odds ratio calculation



1. Case Hours

- Hours switch between light and dark throughout the year
 - **AM: 05:00–07:59**
 - **PM: 17:00–19:59**
- Twilight excluded (solar alt. 0° to -6°)



2. Control Hours

- Hours always light or always dark year-round
- **11:00 PM – 2:00 PM (Next Day)**

Case-Control Method

Case-control design and odds ratio calculation



Odds Ratio Equation ⁽⁸⁾

$$OR = (A / B) / (C / D)$$

A Count in case hour when **light**

B Count in case hour when **dark**

C Count in control hour when **light**

D Count in control hour when **dark**

Pooled Mantel-Haenszel OR ⁽⁸⁾⁽⁹⁾

$$MH^wOR = \frac{\Sigma(A \cdot D / N)}{\Sigma(B \cdot C / N)}$$

A, B, C, D Per-pair 2×2 table cells

N = A + B + C + D

Σ Over all valid hour pairs

[8] Fotios et al. (2024), odds-ratio sensitivity

[9] Mantel, N. and Haenszel, W. (1959) Journal of the National Cancer Institute, 22(4), 719-748.

Generalised Linear Model: Why?

- Outcome = pedestrian counts (discrete and non negative)
- Data is overdispersed (variance > mean)
- Includes: time, weather₍₁₀₎₍₁₁₎, lags, darkness
- Interpretable effects (percentage change)
- Link: $\log(E(\text{ped}))$

(10) Met Office. MIDAS Open: UK hourly weather observation data, v202507. NERC EDS Centre for Environmental Data Analysis, 2025.

(11) Met Office. MIDAS Open: UK hourly rainfall data, v202507. NERC EDS Centre for Environmental Data Analysis, 2025



GLM: Base Model Specification

Model Specification

$$\log\{E(y_{it})\} = \beta_0 + \beta_1 \text{Dark}_{it} + \beta_2 \text{Lag1} + \beta_3 \text{Lag24} + \gamma_h + \delta_d + \mu_m + \theta W_t + \alpha_i$$

y_{it} Hourly pedestrian or cyclist count at sensor i in hour t

$\text{Lag1}, \text{Lag24}$ Count at same sensor 1 h and 24 h ago (captures short-run autocorrelation)

$\gamma_h, \delta_d, \mu_m$ Hour-of-day, day-of-week, and month fixed effects

α_i Sensor fixed effect: absorbs all time-invariant location differences (footfall level, neighbourhood type, infrastructure)

θW_t Temperature, wind speed, rainfall in hour t

Dark_{it} Binary: 1 if sun is below horizon (civil twilight excluded), 0 if daylight

[12] *Negative Binomial Regression: Second Edition*, by Joseph M. Hilbe



GLM: Base Model Components

- Time (hour, day, season) → patterns
- Lag variables → capture short-term autocorrelation
- Weather → affects activity
- Darkness → main variable



GLM: Additional Model Specification

Model Specification

$$\log\{E(y_{it})\} = \beta_0 + \beta_1\text{Dark}_{it} + \beta_2\text{Lag1} + \beta_3\text{Lag24} + \gamma_h + \delta_d + \mu_m + \theta W_t + \alpha_i$$



(13) Bristol City Council. Council cctv cameras. data.gov.uk dataset, 2024.
(14) 2025 Bristol City Council (2025), Quality of Life Survey 2024 to 2025: Final Report
(15) Bristol City Council. Streetlights and street furniture. data.gov.uk dataset, 2024.
(16) OpenStreetMap contributors. Openstreetmap. OpenStreetMap database, 2026

GLM: Comparison with Case-Control

- Case-control: matches light VS dark periods
- GLM: Controls time, weather, autocorrelation...
- Both depict darkness reduces walking

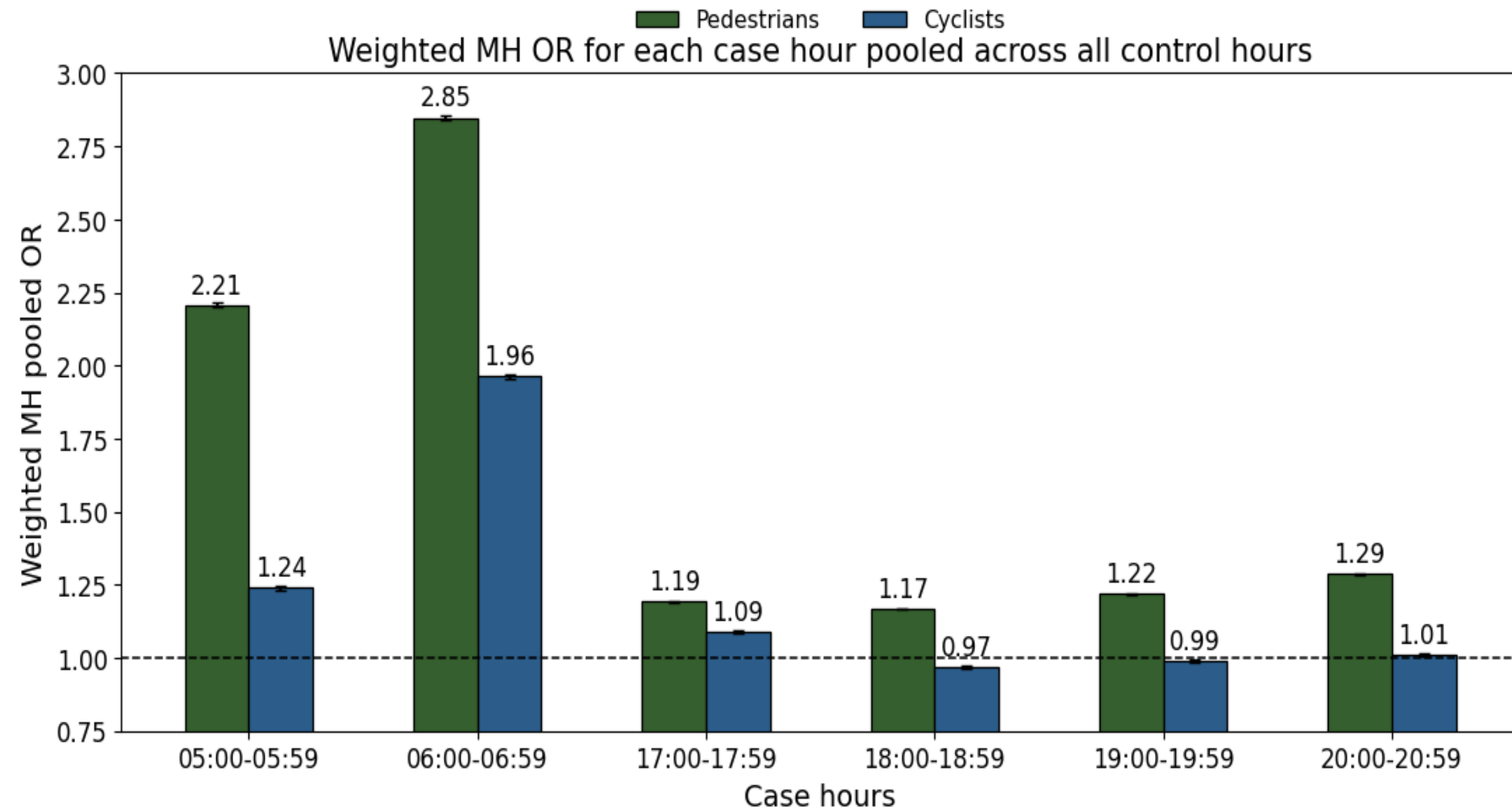


Case-Control Results

Tom

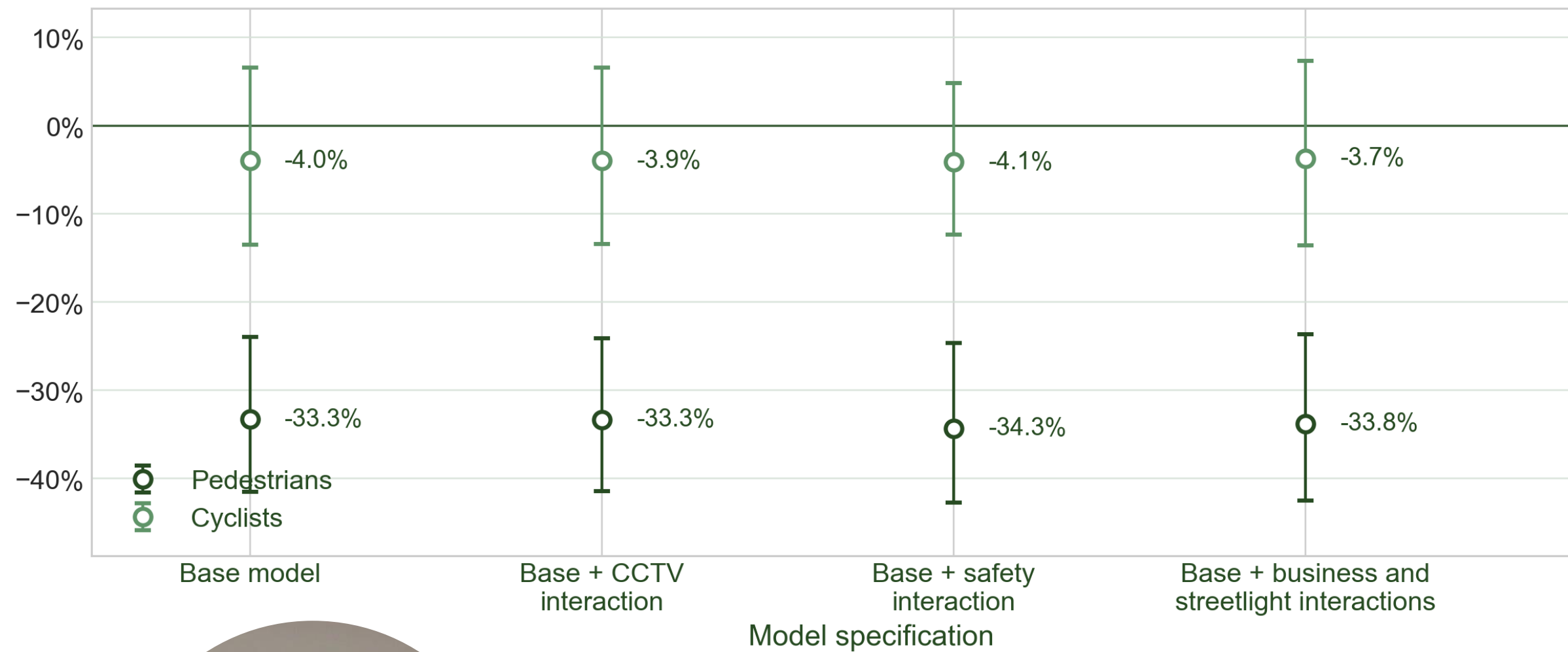


- Odds ratio sensitivity was tested across varying case/control hour combinations
- Largest reduction during morning hours (5am – 7am)
- Darkness suppresses pedestrian counts more than other sensor types



GLM Results

Main darkness effect across model specifications

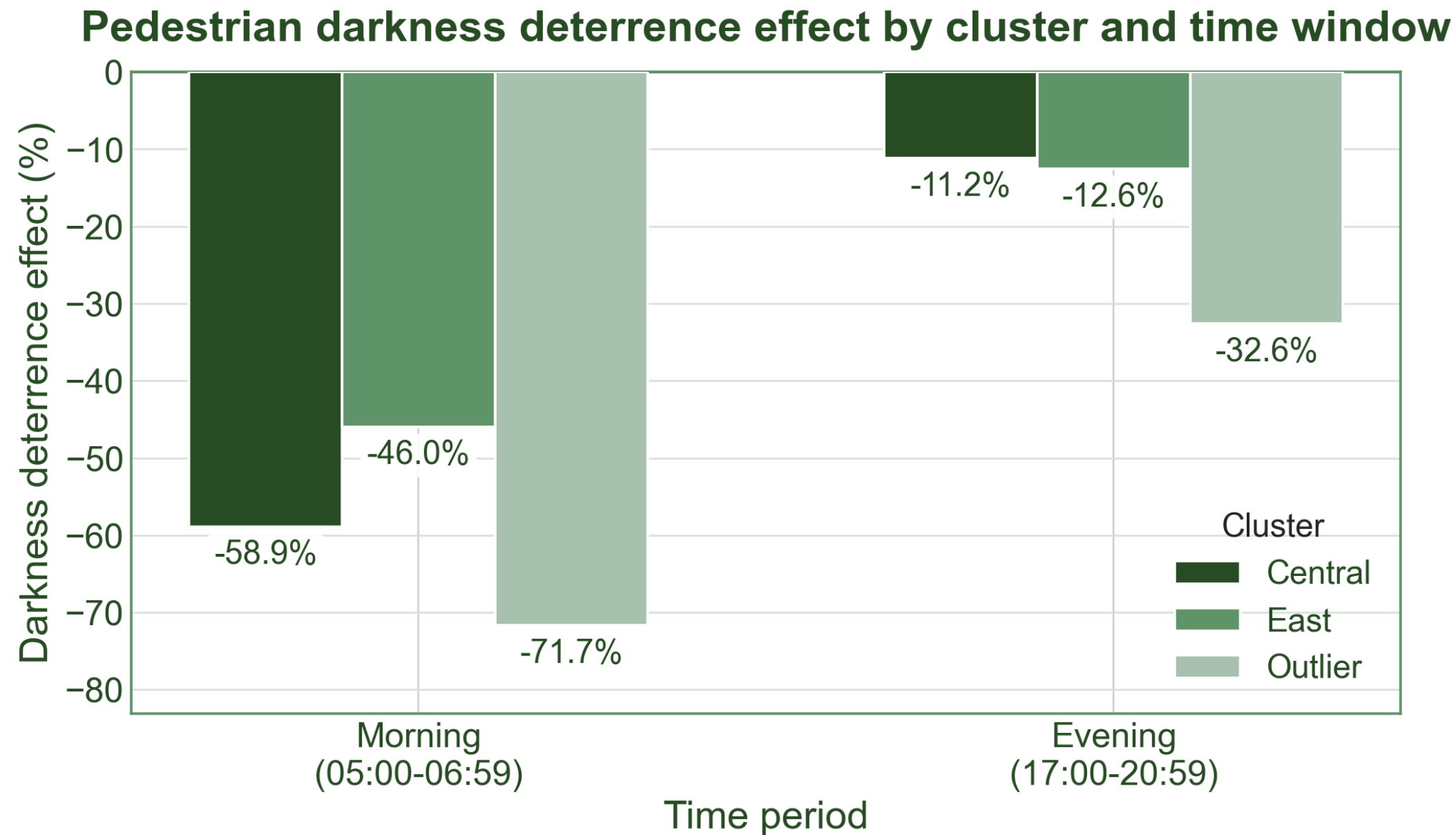


Key Findings

- 97% of sensors show lower pedestrian counts in darkness
- GLM result stable across all model specifications
- P-value for cyclist: 0.313
- P-value for pedestrians: 1.48e-10



Early morning hours need priority action

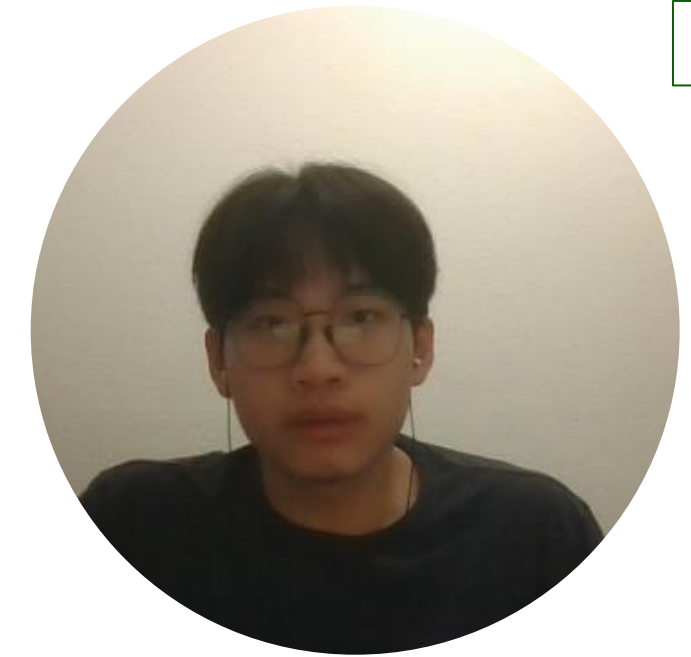


What the time discrepancies show

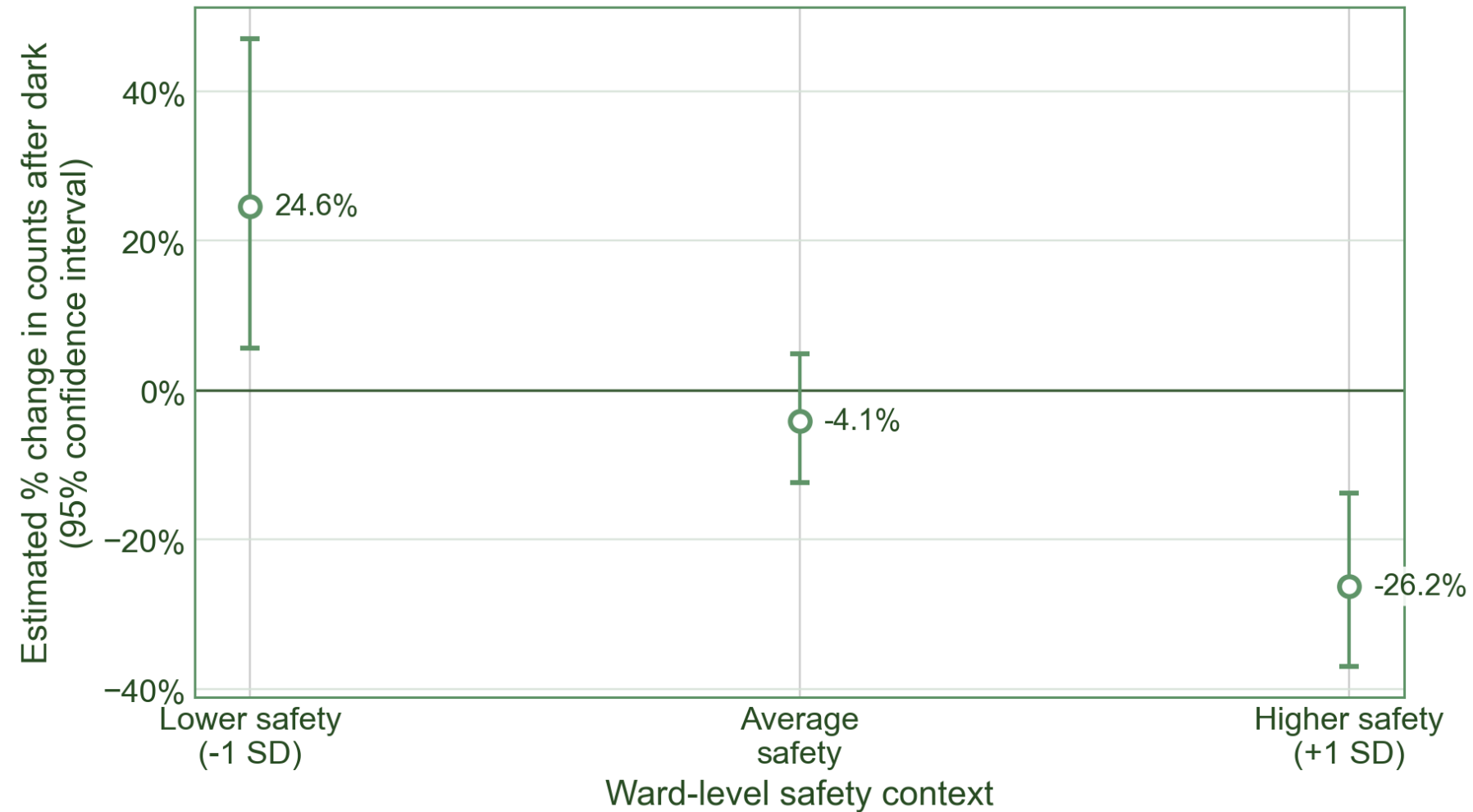
- Morning penalty (–59%) is more than double the evening penalty (–19%)
- Other factors could be the root cause of these differences



Other factors to consider



Cyclist darkness effect by safety context



Moderation Results

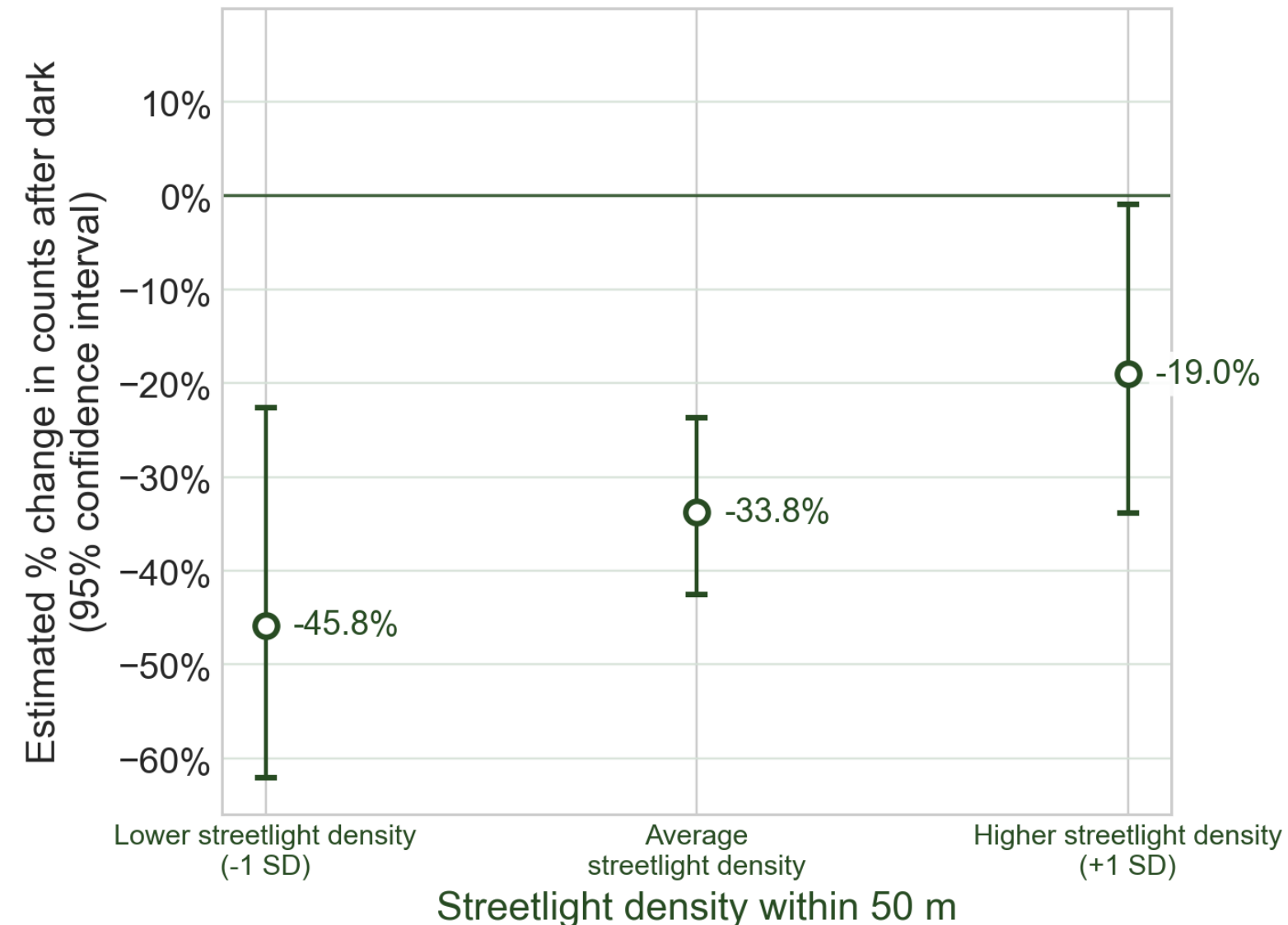
Moderator	Pedestrians	Cyclists
Dark × CCTV	p = 0.948	p = 0.941
Dark × Safety	p = 0.160	p = 0.000121

Key insights:

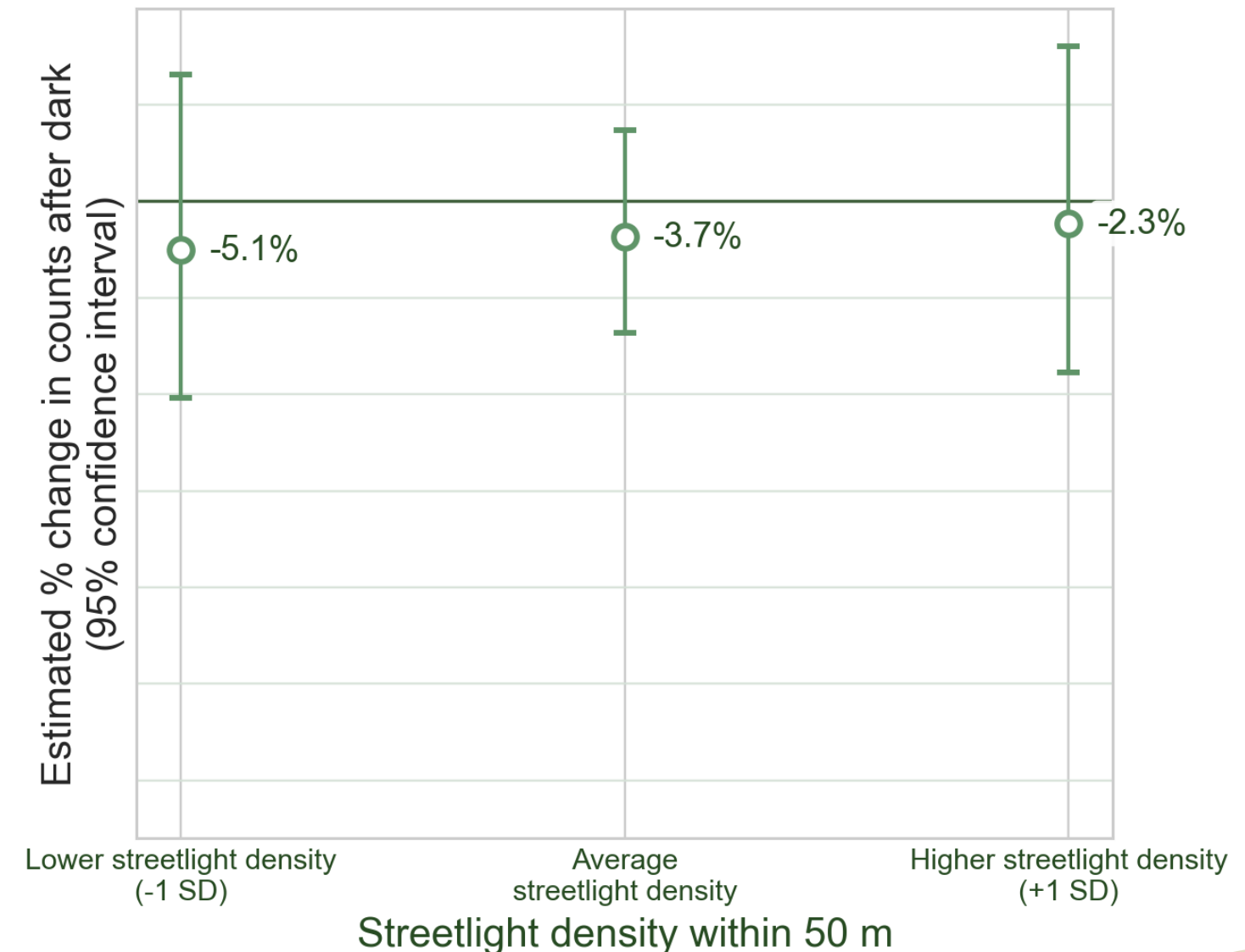
- CCTV has little impact on darkness effect, perhaps down to the location of CCTV being unknown to most people
- Perceived safety by people does have an effect on darkness with a higher ward safety score leading to a larger reduced change in darkness.

Streetlights

Pedestrian darkness effect by nearby streetlight density



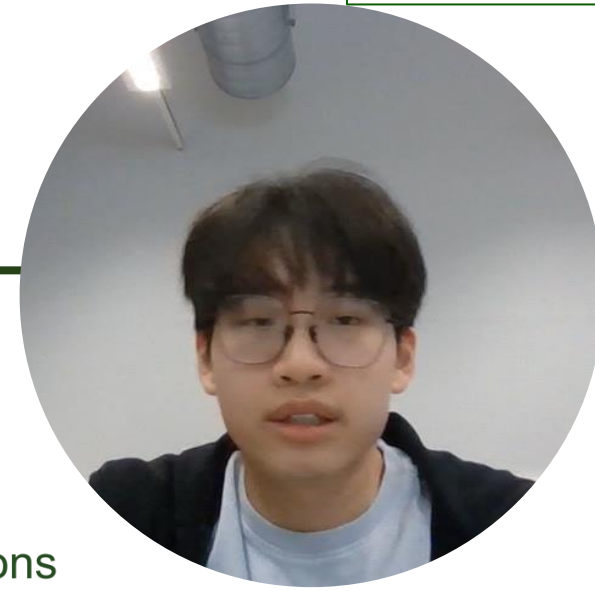
Cyclist darkness effect by nearby streetlight density



- Streetlights cause the largest decrease in the effect of darkness for pedestrians out of all safety factors
- Little to no change for cyclists, likely due to their own lights improving visibility

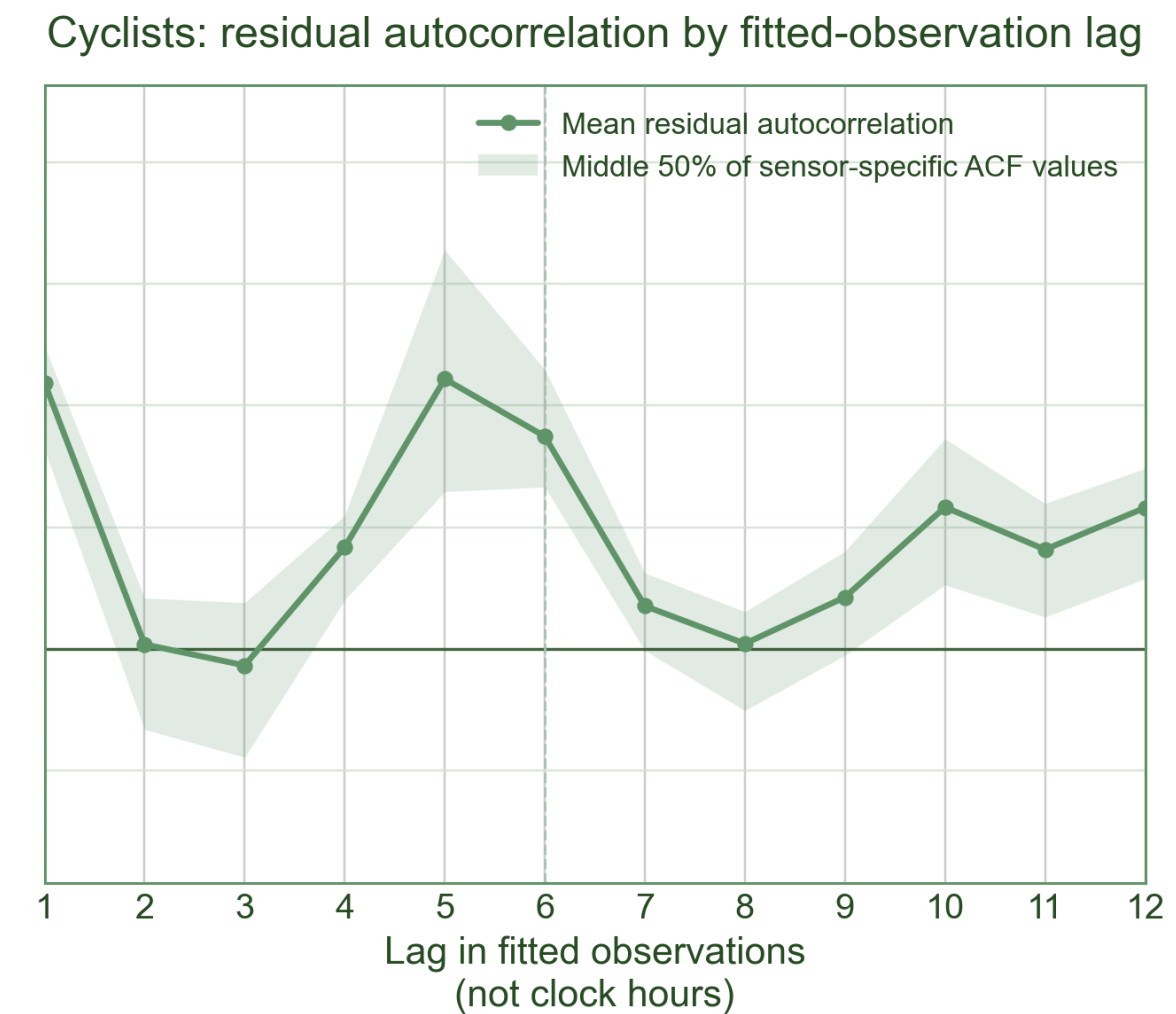
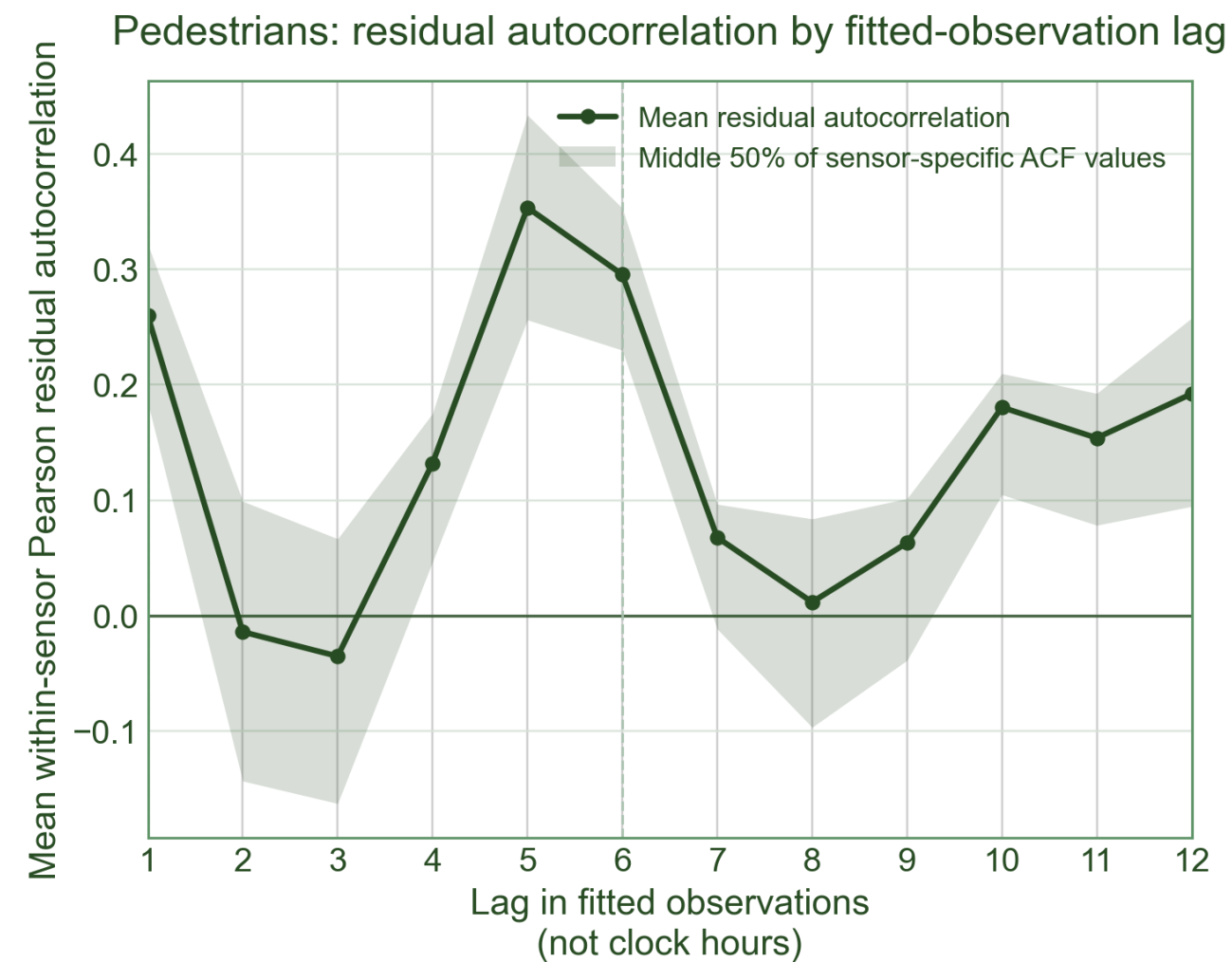


Limitations of the GLM



- Residual autocorrelation, even with the time controls
- Only used case hours, instead of whole time series
- Certain interactions, like darkness with CCTV, are statistically weak

Base model residual autocorrelation across successive fitted observations



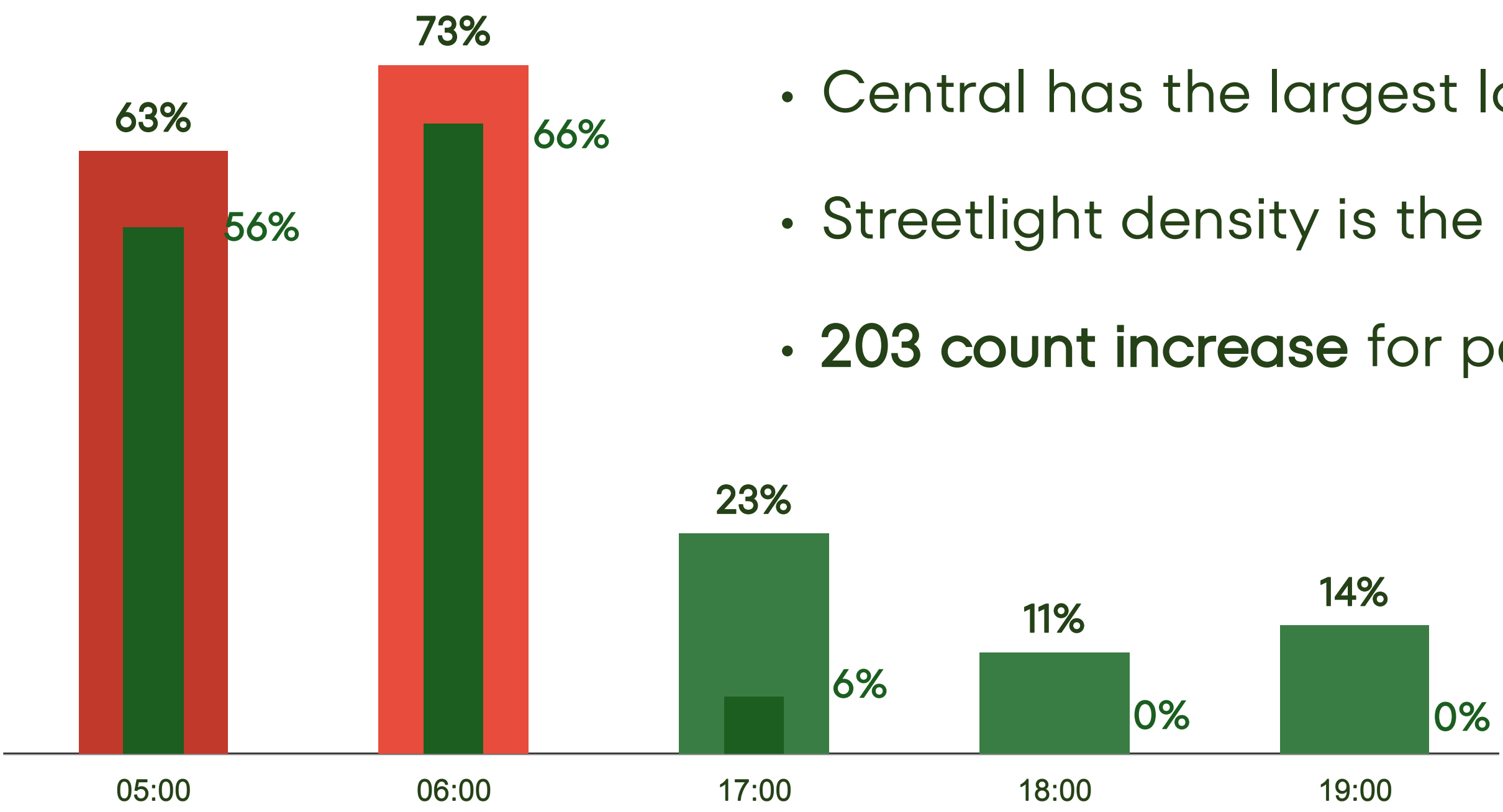
The model is fitted only on 05:00, 06:00, 17:00, 18:00, 19:00, and 20:00, so lag positions refer to successive fitted observations rather than continuous clock-hour gaps.

Targeted Timing: The Morning Commute

MORNING

EVENING

Pedestrian Darkness Penalty by Hour (% Reduction)

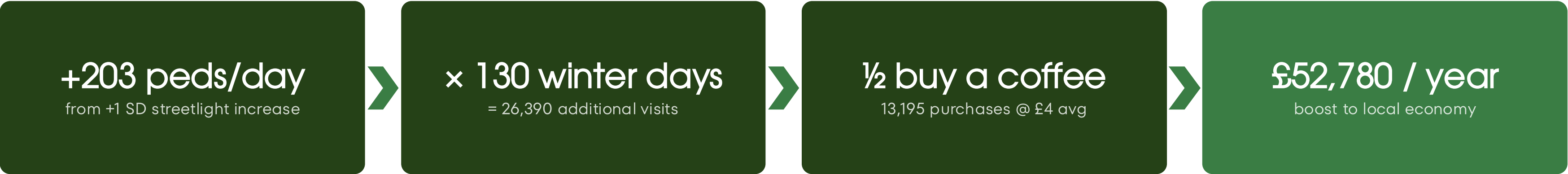


- Central has the largest loss
- Streetlight density is the key factor
- **203 count increase** for pedestrians over **5** hours



LED Streetlight Investment Plan

REVENUE PATH



3.4 Year ROI
£180,250 ÷ £52,780/yr

COST PATH



Recap

1. Gather local data ☒

2. Compare case-control OR and time-series GLM ☒

3. Inform lighting & active-travel decisions ☒



tk24074@bristol.ac.uk